Differentiation & Integration

Section 1: Derivatives of Polynomials & Tangents

1 Differentiate $y = 3x^4 - 5x^2 + 7x - 2$ with respect to x.

a. $12x^3 - 10x + 7$

b. $12x^3 - 10x^2 + 7$

c. $7x^3 - 10x + 7$

d. $12x^4 - 10x^2 + 7$

2 Find the gradient of the curve $y = x^3 - 4x + 1$ at the point where $x = 2$.

a. 8

b. 12

c. 4

d. 0

3 Find the equation of the tangent to the curve $y = x^2 - 3x + 2$ at the point $(3, 2)$.

a. $y = 3x - 7$

b. $y = 3x - 2$

c. $y = x - 1$

d. $y = 2x - 4$

Section 2: Stationary Points & Optimization

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4 Find the stationary points of the curve $y = 2x^3 - 9x^2 + 12x + 1$ and determine their nature.

a. Local max at $(1, 6)$, local min at $(2, 5)$

b. Local min at $(1, 6)$, local max at $(2, 5)$

c. Local max at $(2, 5)$, inflection at $(1, 6)$

d. Local min at $(0, 1)$, local max at $(2, 5)$

5 A metal wire of length 40 cm is bent into a rectangle of dimensions x by y. What value of x maximizes the enclosed area?

a. $x = 10$ cm

b. $x = 20$ cm

c. $x = 5$ cm

d. $x = 15$ cm

6 For what range of values of x is the function $f(x) = x^3 - 6x^2 + 9x - 4$ strictly increasing?

a. $x < 1$ or $x > 3$

b. $1 < x < 3$

c. $x > 2$

d. $x < 2$

Section 3: Indefinite & Definite Integrals

7 Find $\int (6x^2 - 4x + 5)\,dx$.

a. $2x^3 - 2x^2 + 5x + C$

b. $3x^3 - 2x^2 + 5x + C$

c. $2x^3 - 4x^2 + 5x + C$

d. $12x - 4 + C$

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8 Evaluate the definite integral $\int_1^3 (3x^2 - 2x)\,dx$.

a. 18

b. 24

c. 16

d. 20

9 A curve passes through $(1, 4)$ and has gradient function $\frac{dy}{dx} = 4x - 1$. Find the equation of the curve.

a. $y = 2x^2 - x + 3$

b. $y = 2x^2 - x + 4$

c. $y = 4x^2 - x + 1$

d. $y = 2x^2 - x + 1$

Section 4: Area Under Curves & Exam-Style Modeling

10 Find the area enclosed by the curve $y = 4 - x^2$ and the x-axis.

a. $\frac{32}{3}$

b. $\frac{16}{3}$

c. 8

d. $\frac{64}{3}$

11 Find the area of the region bounded by the curve $y = x^2$ and the line $y = 2x$.

a. $\frac{4}{3}$

b. $\frac{2}{3}$

c. 2

d. $\frac{8}{3}$

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12 A particle moves along a straight line such that its velocity at time t is $v(t) = 3t^2 - 12t + 9$. Find the total displacement between $t = 0$ and $t = 3$.

a. 0

b. 9

c. 18

d. 4.5

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